MILITARY SPECIFICATION

FILTER AND UTILITIES UNIT, CHEMICAL-BIOLOGICAL SHELTER SYSTEM, TRAILER MOUNTED: M68

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers that trailer mounted element which provides system control, environmental control, primary power, inflation pressure, electrical energy, and facility for mounting the filters and accessories for the M51 shelter system.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants
QQ-B-654 - Brazing Alloy, Silver.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum, and Aluminum Alloys.
MIL-W-45205 - Welding, Inert-Gas, Metal-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor.
MIL-S-45743 - Soldering, Manual Type, High Reliability, Electrical, Electronic, Instrument, Communication and Radar for Aerospace, and Control Systems Procedures For.
MIL-S-51368 - Shelter System, Collective Protection, Chemical-Biological: Inflatable, 10-Man, Trailer Transported, M51.
MIL-B-55166 - Battery, Storage: Lead Acid, 24 Volt (Starting, Lighting and Ignition - Automotive Type).

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipment (Civilian Agencies).

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

E5-19-5000 - Filters and Utilities Unit, Chemical-Biological Shelter System, Trailer Mounted, M68

PUBLICATIONS

CHEMICAL SYSTEMS LABORATORY TECHNICAL MANUALS

L05-2805-259-12 - Lubrication Order, Engine, Gasoline, 20-HP Military Standard (Model 2A042-II) 10HP FSN 2805-952-3927;
(Model 4A084-II) 20HP FSN 2805-952-3926;
(Model 2A042-III) 10HP FSN 2805-872-5971;
(Model 4A084-III) 20HP FSN 2805-872-5972.

L03-4240-264-12 - Lubrication Order: Collective Protection Shelter System, Chemical-Biological, Trailer Transported: Inflatable, 10-Man, M51.

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TM5-2805-259-14 - Operator, Organization, Direct Support and General Support Maintenance Manual, Engine, Gasoline, 20HP Military Standard.

TM3-4240-264-12 - Shelter System, Collective Protection, Chemical-Biological: M51. (Operator's and Organizational Maintenance)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS, INC.

GUIDE AND DATA BOOK - NCA-60 Curve, Noise Criterion.

(Application for copies should be addressed to the American Society of Heating, Refrigeration, and Air Conditioning Engineers, Inc., 345 East 47th Street, New York, New York 10017).

AIR MOVING AND CONDITIONING ASSOCIATION, INC.

Bulletin No. 210 - Standard Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Moving and Conditioning Association, Inc., 30 West University Drive, Arlington Heights, IL 60004).

3. REQUIREMENTS

3.1 Materials and Components.

3.1.1 Materials. All materials cited on Drawing E5-19-5000 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the Utility Unit shall conform to the specifications and drawings listed on Drawing E5-19-5000 and to subsidiary drawings.

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3.2 Manufacture and assembly. The components shall be manufactured and assembled as specified in the subsidiary drawings listed on Drawing E5-19-5000.

3.3 Performance.

3.3.1 Starting and stopping. The engine of the utility element shall start and stop when tested in accordance with 4.4.4.2.

3.3.1.1 External start. The utility element shall be capable of being started from an external power source using the slave receptacle furnished on the unit, when tested in accordance with 4.4.4.2.1.

3.3.2 Control system.

3.3.2.1 Generator and voltage regulator. The generator frequency shall read 62 (+0, or -4) hertz (hz) and the voltage regulator output shall be 208 ± 10 volts alternating-current (vac) under no load and full load conditions when tested in accordance with 4.4.4.3.1 and 4.4.4.3.4 respectively.

3.3.2.2 DC power supply. The power supply shall supply 26.0 ± 3.0 volts direct-current (vdc) for the controls circuit when tested in accordance with 4.4.4.3.2.

3.3.2.3 Evaporator fan. The evaporator fan shall start and run on 208 vac, 3 phase, 60 hz power when operated in accordance with 4.4.4.3.3.

3.3.2.4 Environmental control unit (ECU) time delay relay. The ECU time delay shall provide a 5 ± 1 second delay between start of the re-circulation fan and the condenser fan when tested in accordance with 4.4.4.3.3.

3.3.2.5 Cooling system. Both the refrigerant compressor and condensing fan shall start and operate to produce refrigeration when controlled as instructed in 4.4.4.3.3.

3.3.2.6 Indicator lights. The indicator lights listed below shall illuminate when tested in accordance with 4.4.4.3.5.

(a) HEATER ON light, auxiliary control panel

(b) LOW OIL PRESSURE light, auxiliary control panel

(c) MASK light, auxiliary control panel

(d) EMERGENCY ENTRANCE MODE light, auxiliary control panel

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3.3.2.7 Shelter and entrance light switch. The shelter and entrance lights shall function when tested in accordance with 4.4.4.3.6.

3.3.2.8 Entrance recirculation fan phase sequence indicator. The ABC lamp on the entrance recirculation fan phase sequence indicator shall light indicating proper three phase wiring to the entrance recirculation fan when tested in accordance with 4.4.4.3.6.

3.3.2.9 Warning light cycling. The warning light shall cycle as shown below when tested in accordance with 4.4.4.3.6.

	Initial Mode	Mode after simulated door opening
Shelter entry light Proceed when on	On	Off
Entrance entry light Do not enter when on	Off	On
Shelter exit light Do not exit when on	Off	On

3.3.3 Noise. Noise inside the shelter shall be determined for vent, cool, and heat modes using applicable instrumentation and shall not exceed the noise criterion curve NCA-60 of the ASHRAE Guide and Data Book when tested in accordance with 4.4.4.4. This test may be performed as part of preproduction testing of MIL-F-51368 for all systems procurements.

3.3.4 Ventilating system.

3.3.4.1 Air leakage. The air leakage of that portion of the ventilating system between the centrifugal blower, the entrance of the return air plenum and outlet of the supply air plenum shall not exceed 10 cubic feet per minute (cfm) at 3.0 inches water gage (iwg) when all deliberate openings have been blocked and sealed and tested in accordance with 4.4.4.7.1.

3.3.4.2 Evaporator fan airflow. Operating in the recirculation mode with both the supply and return ducts connected and with an external resistance of 0.5 $\pm$ 0.1 iwg at the end of the supply duct the fan shall provide 975 cfm minimum at 208 vac, 3 phase, 60 hz power supply when tested in accordance with 4.4.4.7.2.

3.3.5 Refrigerant System.

3.3.5.1 Refrigerant. Type 12 refrigerant, in accordance with BB-F-1421, shall be used.

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3.3.5.2 Leakage. When tested in accordance with 4.4.4.5.1 the entire refrigerant system as assembled in the final configuration shall contain between 8.5 to 9 pounds of refrigerant and withstand pressure up to 400 pounds per square inch gage (psig) for one minute without deformation. Maximum allowable leakage at a pressure equal to or greater than the saturation pressure of type 12 refrigerant at 60°F shall be 0.5 ounce (oz) of refrigerant per year from a single leak (excluding the compressor shaft seal). The maximum allowable leakage at the compressor seal shall be 1.0 oz per year.

3.3.5.3 High pressure cutout switch. The refrigerant high pressure cutout switch shall operate, and stop the compressor, at 300 + 10 psig discharge (condenser) pressure when tested in accordance with 4.4.4.5.2.

3.3.5.4 Low pressure cutout switch. The refrigerant low pressure cutout switch shall operate, and stop the compressor at 20 + 5 psig and restart the compressor when tested in accordance with 4.4.4.5.3.

3.3.6 Heating system.

3.3.6.1 Heater. The heater shall start and produce heat, both high and low range when operated in accordance with 4.4.4.6.1. An increase in temperature shall result when the heater is switched from low range to high range.

3.3.6.2 Heat thermostat. The heater shall cycle and also not cycle when the air conditioning switch on the auxiliary control panel is set on HEAT and tested in accordance with 4.4.4.6.2.

3.3.6.3 Flame detector switch. The heater igniter shall meet the following conditions when tested in accordance with 4.4.4.6.3:

(a) Heater igniter shall function for up to 2 minutes unless flame detector switch changes to full heat (either high or low heat position).

(b) Heat igniter shall signal "Heat" by the heater or indicator lamp on the auxiliary control panel.

3.3.6.4 Ignition time cutoff. The heater shall attempt to fire for 4-1/2 ± 1/2 minutes before cutoff of safety relay after failure to fire when tested in accordance with 4.4.4.6.4.

3.3.7 Pressurization system. The centrifugal blower and pressure control system shall be capable of maintaining 0.3 iwg positive pressure in the inlet compartment of the ECU ventilating system when the delivered makeup air is 160 cfm maximum and tested in accordance with 4.4.4.8.

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3.3.7.1 Makeup air pressure switch. The makeup air pressure switch shall open and cause the solenoid valve to open on a falling pressure at 0.3 ± 0.1 iwg when tested as indicated in 4.4.4.8.1.

3.3.7.2 Low pressure warning switch. The shelter low pressure warning switch (mask light switch) shall close on a falling pressure at 0.15 ± 0.1 iwg, and operate a warning buzzer and MASK indicator light when tested in accordance with 4.4.4.8.2.

3.3.8 Service life. The unit shall operate satisfactorily for 500 hours (400 hours on cooling cycle, 100 hours on heating cycle) with no failures. This test will be performed as part of preproduction and reliability qualification of MIL-S-51368 for all systems procurement. Follow-on procurement for utility units only shall require a test in accordance with 4.4.4.9.

3.3.9 Operational. Each unit shall be operated a total of 24 hours at the generator speed producing 62 (+0, or -4) hz electric current. The operating cycle shall consist of one hour of heating, one hour of cooling, and one hour of recirculation. The ventilating system shall operate under a positive pressure of 1.0 ± 0.1 iwg, when tested in accordance with 4.4.4.10.

3.4 Preproduction. Prior to start of regular production, a preproduction sample shall be produced in accordance with this specification, for examination and test.

3.5 Workmanship.

3.5.1 Metal fabrication. Sheet metal and structural members used in fabrication shall be straight. Sheet metal and structural members may be cut by shearing or sawing. All burrs and rough edges shall be removed. Corners shall be square and true. All bends shall be made to a radius which will not cause cracking at the extreme fibers, and all bends except for minor details shall be made with metal dies or fixtures to ensure uniformity in size and shape.

3.5.2 Bolted connections. Bolt holes shall be punched or drilled and all burrs removed. Washers and lockwashers shall be provided and all bolts, nuts, and screws shall be tightened to prevent loosening under vibration.

3.5.3 Riveted connections. Rivet holes shall be punched or drilled and all burrs removed. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the members.

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3.5.4 Welding. The surfaces of parts to be welded shall be free of oxide, scale, paint, grease, oil, or any other foreign matter. Immediately before welding, oxide shall be removed by an abrading or a chemical process. Grease and oil shall be removed by wiping with a solvent such as acetone. All surfaces shall be dried after cleaning. Welds shall transmit stress without permanent deformation when subjected to proof and service loadings. All welded parts shall be free of cracks and other imperfections that may reduce the effectiveness of the parts. Inert gas shielded arc welding shall be used for aluminum welds in accordance with MIL-W-45205 for class A weldments.

3.5.5 Soldering. Soldering of all electrical connections shall be in accordance with MIL-S-45743.

3.5.6 Brazing. Refrigerant line brazing shall conform to the requirements of MIL-B-7883 using silver solder in accordance with QQ-B-654, classes III and IV. During the brazing operation, a small amount of suitable gas, such as dry nitrogen, shall be continuously bled through the refrigerant line to prevent oxidation within the line. Flux shall not be present in the refrigerant system, and heavy beads or fillets of brazing material shall be avoided.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.1 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction.

4.3.1 Sample. A sample of two M68 filters and utilities units shall be produced, using the same methods, materials, and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The two units shall be examined in accordance with 4.4.3.3.

4.3.2.2 For tests. One utility element selected by the Government shall be subjected to all the tests in accordance with table I.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified in 4.3.2.1 and 4.3.2.2 to be acceptable. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of one utility element produced by one manufacturer, at one plant from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Each trailer shall be examined and tested.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Each utility unit shall be examined in accordance with the list of defects (4.4.3.3). Any defect shall be cause to reject the unit.

4.4.3.2 For test. Each utility unit shall be tested for quality conformance as specified in table I and in accordance with the applicable tests specified in 4.4.4. Any unit that fails any test shall be rejected.

4.4.3.3 List of defects.

101	Low refrigerant charge
102	Parts or components missing
*103	Installation of wiring not as specified

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4.4.3.3 List of defects (continued).

104	Components not lubricated
105	Treatment and painting defects such as blisters, thin spots, lumps, scratches, etc.
106	Nameplates and stenciling not as specified
107	Technical publications missing
108	Evaporator and condenser coil fins bent other than as required
*109	Metal fabrication not as specified (3.5.1)
*110	Bolted or riveted connections not as specified or missing (3.5.2 and 3.5.3)
*111	Welding defects (3.5.4)
*112	Soldering and brazing (3.5.5 and 3.5.6)
113	Fasteners that do not function as intended
114	Evidence of fuel or oil leakage

* To be checked in-process.

4.4.4 Tests. All tests shall be conducted in an ambient temperature between 60°F to 90°F unless otherwise specified herein. When performing tests of a M68 utilities unit the following items which are not part of the M68 unit, shall be available at the time of test:

Dwg/Part No.	Item*
E5-19-6565	Inspection Aid
E5-19-5008	Control-Indicator (Auxiliary)
C5-19-5090	Cable Assembly, Power, Electrical
D5-19-5091	Fixture, Lighting
D5-19-5095	Fixture, Lighting
D5-19-1221	Filter, M23
D5-19-1226	Filter, M24
C5-19-5209-10	Filter, Particulate
E5-19-5211-10	Filter, Gas
E5-19-5154	Plenum, Sound-Attenuating
E5-19-5063	Plenum, Box
D5-19-5060-100	Air Duct Assly, 12 in., Shelter (Return)
D5-19-5060-101	Air Duct Assly, 12 in., Shelter (Supply)

*Equivalents may be substituted for these items after Government approval.

Definition of failure. Performance deviations from the requirements set forth in 3.3 shall constitute a failure. Any failures of test equipment or tooling during these tests are not accountable failures. No equipment performance deviations shall be permitted.

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TABLE I. Inspection schedule

Inspection sequence	Type of inspection: Preproduction	Type of inspection: Quality conformance	Examination or test	Requirement paragraph	Inspection paragraph
1	2	3	4	5	6
1	X	X	Examination		4.4.3.3
2	X	X	Refrigerant leakage	3.3.5.2	4.4.4.5.1
3	X		Starting and stopping	3.3.1	4.4.4.2
4	X		External start	3.3.1.1	4.4.4.2.1
5	X		Generator and voltage regulator (no load)	3.3.2.1	4.4.4.3.1
6	X	X	DC power supply	3.3.2.2	4.4.4.3.2
7	X	X	Evaporator fan	3.3.2.3	4.4.4.3.3
8	X	X	ECU time delay relay	3.3.2.4	4.4.4.3.3
9	X	X	Cooling system control	3.3.2.5	4.4.4.3.3
10	X		Generator and voltage regulator (full load)	3.3.2.1	4.4.4.3.4
11	X	X	Indicator lights	3.3.2.6	4.4.4.3.5
12	X	X	Shelter light switch	3.3.2.7	4.4.4.3.6
13	X	X	Entrance fan indicator	3.3.2.8	4.4.4.3.6
14	X	X	Warning light cycle	3.3.2.9	4.4.4.3.6
15	X		Noise	3.3.3	4.4.4.4

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TABLE I. Inspection schedule (continued)

Inspection sequence	Type of inspection: Preproduction	Type of inspection: Quality conformance	Examination or test	Requirement paragraph	Inspection paragraph
1	2	3	4	5	6
16	X	X	High pressure cutout	3.3.5.3	4.4.4.5.2
17	X	X	Low pressure cutout	3.3.5.4	4.4.4.5.3
18	X		Air leakage	3.3.4.1	4.4.4.7.1
19	X		Evaporator fan airflow	3.3.4.2	4.4.4.7.2
20	X	X	Heating system starting	3.3.6.1	4.4.4.6.1
21	X	X	Thermostat	3.3.6.2	4.4.4.6.2
22	X	X	Flame detector	3.3.6.3	4.4.4.6.3
23	X	X	Ignition time cutoff	3.3.6.4	4.4.4.6.4
24	X		Pressurization system	3.3.7	4.4.4.8
25	X	X	Makeup air pressure switch	3.3.7.1	4.4.4.8.1
26	X	X	Low pressure warning switch	3.3.7.2	4.4.4.8.2
27	X*		Service life	3.3.8	4.4.4.9
28	X	X	Operational	3.3.9	4.4.4.10

*To be performed as part of tests required by the Preproduction and reliability qualification test of MIL-S-51368 during system procurements. In the event of a procurement of only utility units the service tests shall be performed using guidelines of Preproduction and reliability qualification test of MIL-S-51368.

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4.4.4.1 Pretest operations.

4.4.4.1.1 Engine. Prior to and throughout the tests, the engine shall be maintained and lubricated in accordance with TM5-2805-259-14 and LO5-2805-259-12.

4.4.4.1.2 Centrifugal blower. Prior to starting the engine, the centrifugal blower shall be filled with tractive fluid in accordance with LO3-4240-264-12.

4.4.4.1.3 Belt tensions. Prior to starting the gasoline engine, the drive belts shall be checked for the following deflections by means of a straight edge and tensiometer.

(a) Refrigerant compressor belts shall require 4 (+ 1, or -0) pounds (lbs) load for a 5/16 inch deflection.

(b) Inflation centrifugal blower belt shall require 4 (+ 1, or -0) lbs load for a 1/8 inch deflection.

(c) The electric generator belt shall require 4 (+ 1, or -0) lbs load for a 3/16 inch deflection.

4.4.4.2 Starting and stopping. Perform pretest operations. Test the utility element for starting and stopping in accordance with the startup and shutdown procedures contained in TM3-4240-264-12.

4.4.4.2.1 External start. Utilizing a 24-volt military standard battery (MS 75047(AT) or equivalent), and following the startup instructions on the main control panel cover, start the utility unit using external power supplied through the slave receptacle. A standard slave cable or a similar cable shall be used to perform this test.

4.4.4.3 Control system.

4.4.4.3.1 Generator and voltage regulator (no load).

(a) Remove top cover of main control panel.

(b) With the engine operating and the throttle control in the full forward (in) position, place the AUXILIARY 120 VAC circuit breaker on the main control panel to ON. The voltage measured at the three top terminals on CB-2 (3 phase) shall be 208 ± 10 vac between phases and the steady state frequency reading shall be 62 (+0, or -4) hz.

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4.4.4.3.2 DC power supply. With the utility unit operating in accordance with 4.4.4.3.4, measure the voltage at the terminals of the control circuit fuse. Value of 3.3.2.2 shall be met.

4.4.4.3.3 Evaporator fan, ECU time delay relay, and cooling system control. With the utility element operating per 4.4.4.3.1, connect the auxiliary control panel and place the AIR CONDITIONING switch to COOL. Position the CONDENSER FAN and RECIRCULATION FAN circuit breaker to ON simultaneously and note the delay between the start of the evaporator fan and the condenser fan. Requirements of 3.3.2.3, 3.3.2.4, and 3.3.2.5 shall be met.

4.4.4.3.4 Generator and voltage regulator (full load). With the utility unit operating as in 4.4.4.3.3, measure voltage at the top three terminals on CB-2. Voltage shall be 208 (+ 10) vac between phases and the steady state frequency reading shall be 62 (+0, or-4) hz.

4.4.4.3.5 Indicator light. With unit operating, AUXILIARY 120 VAC circuit breaker engaged and auxiliary control panel properly connected, depress face of HEATER ON, LOW OIL PRESSURE, MASK AND EMERGENCY ENTRANCE MODE lights. Indications shall conform to 3.3.2.6.

4.4.4.3.6 Sheltered entrance lights, entrance fan phase sequence indicator, and warning light cycling. Connect the auxiliary control panel to the distribution box cable (D5-19-5090) and connect this cable to the inspection aid (E5-19-6565). Also connect the shelter light cable (D5-19-5091) and the entrance light cable (D5-19-5095) to the inspection aid. Connect an M68 filter and utilities unit to the auxiliary control panel and start M68. Engage the AUXILIARY 120 VAC and RECIRCULATION FAN AND ENTRANCE FAN circuit breakers on the main control panel. Operate the LIGHTS switch on the auxiliary control panel. Lights shall function as specified in 3.3.2.7. The phase sequence indicator for the entrance recirculation fan on the inspection aid shall function as specified in 3.3.2.8. Prior to conducting the following test wait at least five minutes after engaging circuit breakers. Place the emergency entry mode switch on the auxiliary control panel in the 5 minute mode. Push entrance entry switch on inspection aid simulating entrance door opening. The warning lights shall function as specified in 3.3.2.9.

4.4.4.4 Noise. Noise shall be determined inside the shelter at the geometric center at a height of 5 feet from the floor. This test shall be conducted in the vent, cool, and heat modes and shall meet the requirements of 3.3.3.2.

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4.4.4.5 Refrigerant system.

4.4.4.5.1 Refrigerant leakage. The system will be charged to 25 psig minimum with R12, then increased to 400 psig with nitrogen, and this pressure held for 1 minute minimum. Halogen and/or soap bubble leak tests shall be performed. Leaks shall be repaired as noted. The system charge shall then be vented to atmosphere and the refrigerant system evacuated to 500 microns Hg. maximum. An R12 gas charge at a pressure equivalent to the prevailing ambient temperature shall then be introduced to break the system vacuum. The system shall be examined using a leak detector set to a sensitivity of 0.5 ounces per year. No system leaks in excess of this value shall be accepted with exception of the compressor shaft seal whose acceptable leakage rate is 1.0 ounces per year. At the completion of this test, the refrigeration system will be charged with 8.5 (+0.5, -0.0) pounds of R-12. This weight is to include the initial gas charge.

4.4.4.5.2 High pressure cutout. Install a set of calibrated refrigerant gages on the suction and discharge service valves. Set the air conditioning switch on the auxiliary control panel to COOL with the unit operating. Partially block the condenser airflow and record pressure at which compressor stops. Failure to meet the requirements of 3.3.5.3 constitutes failure of test.

4.4.4.5.3 Low pressure cutout. Under conditions as above requiring cooling, but without condenser air restricted, slowly close the suction isolation valve before the compressor causing a drop in the suction (or low) gage pressure. Record the pressure at which the compressor stops. Caution: Do not completely close the low pressure isolation valve. Increase the low pressure isolation valve setting, and ensure the compressor restarts. Failure to meet the requirements of 3.3.5.4 constitute failure of this test.

4.4.4.6 Heating system.

4.4.4.6.1 Starting. Locate a temperature measuring device (thermometer or thermocouple) in the outlet plenum. With the unit operating, place the AIR CONDITIONING switch on the auxiliary control panel in the HEAT position and HEATER OUTPUT switch in the LOW position. Note and record the time elapsed before the HEATER ON indicator lamp on the panel comes on. Record outlet plenum temperature. Switch the HEATER OUTPUT switch to HI position. Note and record any change in heat output. Failure to meet 3.3.6.1 requirements constitutes failure of test.

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4.4.4.6.2 Thermostat. Remove the top access panel on the main control panel and rotate the set point adjustment knob on the thermostat clockwise to the limit. With the AIR CONDITIONING switch in the HEAT position, observe that the heater does not attempt a start. Failure to meet 3.3.6.2 requirements constitutes failure of test. Rotate set point knob slowly counter-clockwise, and observe setting at which heater starts. Reset knob to original setting. Observe that the heater automatically cycles by placing the sensing element alternately in cold (approximately 50°F) water then in hot water (approximately 100°F) at least ten times.

4.4.4.6.3 Flame detector switch. With the unit operating, place the AIR CONDITIONING switch on the auxiliary control panel in the HEAT position, note and record the elapsed time after closing the switch until the HEATER ON indicator light comes on. Failure to meet the requirements of 3.3.6.3 constitutes failure of test.

4.4.4.6.4 Ignition time cutoff. Disconnect the electric power supply of the heater fuel pump at the fuel pump RFI filter connector. With the unit operating, place the AIR CONDITIONING switch in the HEAT position, observe and record the elapsed time during which heater attempts to fire. Failure to meet 3.3.6.4 requirements constitutes failure of test. Properly reconnect electric supply to fuel pump.

4.4.4.7 Ventilating system.

4.4.4.7.1 Air leakage test. Block and seal the shelter inflation outlet, the inlet of the return air plenum and the outlet of the air supply plenum. Disconnect the centrifugal blower from the makeup filter plenum and connect an air supply capable of maintaining 10 cfm airflow (in series with an airflow measuring device approved by AMCA Bulletin 210) against a 3 iwg pressure in the inlet compartment of the ECU ventilating system. A measured airflow in excess of requirement 3.3.4.1 shall constitute failure of test.

4.4.4.7.2 Evaporator fan air flow. The evaporator fan capacity shall be checked while the plenum, recirculation filters and ducts are in place and the external pressure at the end of the supply duct is 0.5 $\pm$ 0.1 iwg. This test shall be made in accordance with the requirements of AMCA bulletin no. 210. This test shall be considered failed if the fan does not duplicate the capacity specified in 3.3.4.2.

4.4.4.8 Pressurization system. The recirculation system shall be formed into a closed loop system through the use of external ducting. Establish a 1.0 iwg pressure in the external ducting by adjusting the leakage in this ducting to 110 cfm minimum. There shall be a 0.3 inch minimum water gage positive pressure in the inlet compartment of the ECU as specified in 3.3.7 under these conditions.

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4.4.4.8.1 Makeup air switch and solenoid valve test. Slowly reduce the inlet compartment pressure by regulating the damper until the pressure switch opens and opens the solenoid makeup air valve: this point can be determined by an audible functioning of the solenoid on a falling pressure. Inability to meet the requirements of 3.3.7.1 constitutes failure.

4.4.4.8.2 Low pressure warning switch. With the unit operating as in the makeup air switch and solenoid test (4.4.4.8.1), connect the auxiliary control panel. Continue to slowly drop the pressure in the ECU inlet compartment below 0.15 iwg. Observe the functioning of MASK light and buzzer. Requirements of 3.3.7.2 shall be met.

4.4.4.9 Service life test. All operating hours employed in unit testing shall be considered part of the service life test (reference paragraph 3.3.8). After all other testing has been completed, any additional hours required to make up the hour total test time shall be run.

4.4.4.10 Operational. After all functional tests have been completed, each unit shall have an operational test of 24 hours, with the auxiliary control panel and shelter lights properly connected. The supply air and return air ducts shall be connected to a room or chamber of not less than 1200 cubic feet capable of maintaining a positive pressure of 1 iwg with a 0 to 150 cfm controllable leakage. The auxiliary control panel may be mounted outside the chamber for convenience. Means of recording dry bulb temperatures before the evaporator and after the heater shall be established. Temperature at time of change from cooling to heating to recirculation to cooling shall be taken and recorded on a suitable form. The heat transfer in the cooling system shall be checked by measuring the dry bulb temperature before and after the evaporator during the cooling cycles of the operational test. Temperature shall be measured while the compressor is operating and the compressor shall have operated continuously for a minimum of 1 minute prior to the temperature measurement. The dry bulb temperature difference shall be a minimum of 5°F. The thermostat setting may be changed to insure air conditioner and heater operation. All malfunctions or deficiencies apparent during this test shall be corrected. Any malfunction occurring after the first 12 hours of test shall, at the option of the Government the unit shall be required to operate 12 hours, without malfunction, after correction of the original fault. Inflation outlet must be kept capped during this test. Requirements of 3.3.9 shall be met.

5. PACKAGING

5.1 Interplant shipment (see 6.3). The gas and particulate filter (C5-19-5209-10, C5-19-5211-10, D5-19-1221 and C5-19-1226) packaged in accordance with their packaging data sheets and the utilities unit, less filters (E5-19-5000) shall be packed to prevent damage in shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations applicable to the mode of transportation.

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5.2 Repair parts. When the M68 filter and utilities unit is procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on its packaging data sheets which is identified by its National stock number.

5.3 Marking (Interplant shipment). In addition to any special marking required by the contract or order, marking shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The modified M105A2 1-1/2 ton trailer covered by this specification is the trailer facility portion of the M51 shelter system (MIL-S-51368) and is used to house the environmental, inflation, and electrical systems. The trailer also provides mounting points for the M51 shelter and airlock in the traveling mode.

6.1.2 Precautionary notes. To preclude interferences in installation of equipments into the M105A2 trailer, the bodies of all trailers intended for use in this utility element should be checked for the following:

(a) Uniform height of sides, front, and tailgate.
(b) Rectangular integrity of sides to floor, sides to front, and sides to tailgate.
(c) Correct tolerance on dimension between wheel wells.
(d) Conformance to tolerances of trailer (M105A2) drawings.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Interplant shipment. Packaging for supplies and materials that does not enter the military supply system. Typical interplant shipments are shipments from the subcontractor to prime contractor, vendor to a contractor, or a contractor to a military arsenal or plant.

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6.4 International interest. Certain provisions of this specification are the subject of International Standardization Agreement QSTAG 451. When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices if required.

Custodian:

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Preparing activity:

Army - EA

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